LumiGRAD

AI-Powered Academic Performance Intelligence System

Microsoft Track Submission

Our Idea

LumiGRAD is an AI-powered academic intelligence system built to support students continuously, personally, and transparently throughout a semester.

University is very different from high school — especially for freshmen. Suddenly, you're managing modules, projects, internships, hackathons, networking… and no one really tells you what you're actually good at or where you should focus.

And learning today doesn't stop in the classroom. It extends into research, side projects, competitions, and industry opportunities.

But here's the problem:

- How do students apply for opportunities if they don't know their strengths?
- Even if they *do* know, how do they stay in the loop?
- And how do they know whether they're on track academically before it's too late?

That's where LumiGRAD comes in. Instead of just tracking grades, LumiGRAD uses AI agents to:

- Generate active-recall quizzes from lecture content
- Diagnose weak and strong topics
- Explain mistakes clearly
- Track learning states over time
- Estimate GPA readiness
- Recommend targeted reinforcement resources

But we don't stop there.

We turn academic insights into **real-world action**.

With a modern **"For You" page**, inspired by today's apps, students get:

- A breakdown of their strengths and areas to improve
- Research professors aligned with their strong topics
- Relevant job listings and opportunities
- A weekly study plan tailored to their mastery gaps

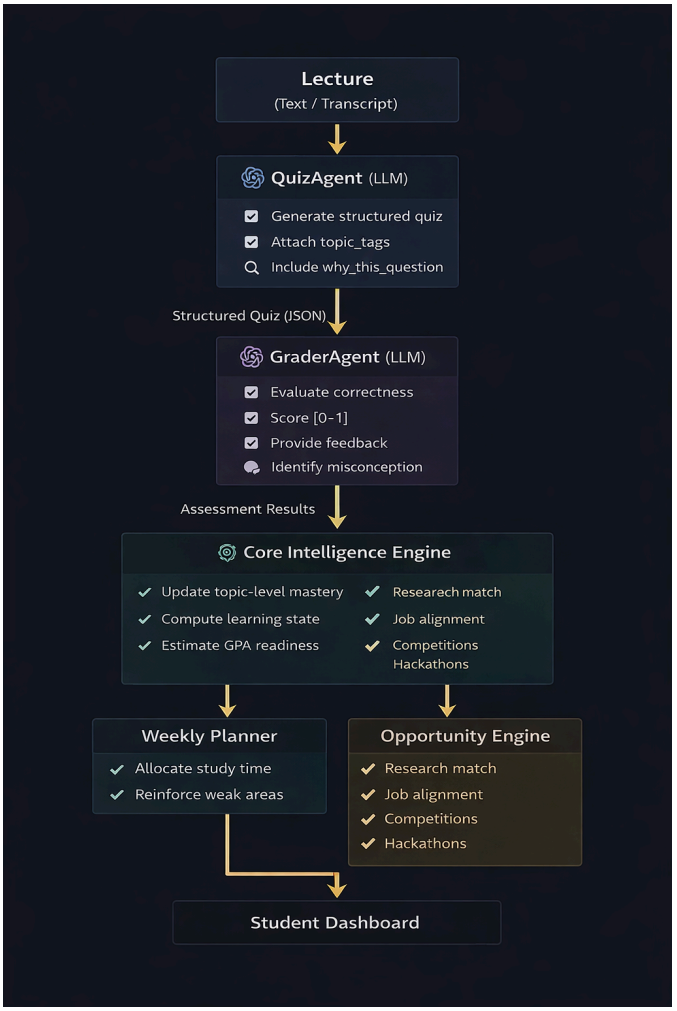
So instead of guessing what to work on — or discovering weaknesses during finals — students get continuous, explainable, and actionable guidance.

LumiGRAD isn't just a tracker.

It's an academic intelligence layer that helps students understand:

- Where they stand
- Where they're improving
- And where they can go next

System Architecture



Flow Explained

1. The student selects the lecture.
2. QuizAgent generates structured quiz
3. GraderAgent evaluates responses.
4. Mastery engine updates topic-level performance.
5. ResourceAgent provides reinforcement materials.

6. Weekly planner adapts schedule.
7. Learning state and GPA risk updated.
8. Opportunity engine aligns strengths to research and job signals.

This creates a continuous feedback loop:

Lecture → Quiz → Diagnosis → Reinforcement → Planning → Monitoring → Alignment

Tech Stack

Frontend

- Streamlit (Interactive Python web framework)

Backend

- Python (Modular architecture)
- REST-based API calls to Azure OpenAI

Database

- SQLite3 (Lightweight relational database for mastery, quiz history, lecture tracking)

AI Layer

- Azure OpenAI (LLM-based agents: QuizAgent, GraderAgent, ResourceAgent)

AI System Design

LumiGRAD uses a modular multi-agent architecture.

QuizAgent

Purpose
QuizAgent generates personalised active-recall questions strictly grounded in lecture content. Its goal is to transform passive lecture material into structured, mastery-driven assessment.

Design Constraints
Each quiz contains approximately 70% multiple-choice questions and 30% short-answer questions. Every question must include topic_tags, why_this_question, and difficulty metadata. The output is strictly formatted in JSON to ensure schema consistency and downstream processing reliability.

Design Principle
The model operates at a temperature of 0.2 to prioritise determinism and educational stability over creative variation. The system prompt enforces lecture grounding, prohibits

hallucinated content, mandates an exact question count, and enforces a structured output schema. This ensures repeatable, explainable, and curriculum-aligned assessments.

GraderAgent

Purpose
GraderAgent performs strict, explainable evaluation of student responses to support conceptual understanding rather than simple correctness scoring.

Design Constraints
Each evaluation returns is_correct (boolean), score_0_to_1 (continuous mastery score), detailed feedback, and identified common_misconception. The output is structured to support analytics and mastery updates.

Design Principle
For every incorrect response, the system must explain the correct concept and explicitly identify the misunderstanding pattern. This shifts the system from binary grading to conceptual correction, reinforcing learning rather than penalising mistakes.

ResourceAgent

Purpose
ResourceAgent provides targeted reinforcement materials for weak topics identified by the Core Intelligence Engine.

Design Constraints
Outputs include structured JSON containing YouTube educational links, academic paper links, and scholar-style search queries. All outputs must remain topic-aligned and lecture-consistent.

Design Principle
When exact authoritative URLs cannot be confidently retrieved, the system falls back to well-formed academic search queries to maintain reliability and avoid hallucinated references. This prioritises trustworthiness and responsible AI behaviour over speculative link generation.

Mastery & Learning State Modeling

Topic mastery is updated incrementally after each graded question using a confidence-aware learning rule. The model integrates three signals: accuracy, attempt confidence, and recency decay.

Accuracy reflects the proportion of correct responses for a topic. Attempt confidence increases as more evidence is collected, reducing sensitivity to small sample sizes. Recency decay applies exponential forgetting when a topic is not practised for multiple days, modeling natural memory decline.

Mastery updates follow a bounded learning rule:

- Mastery shifts toward the latest performance outcome
- Learning rate shrinks as confidence grows
- Exponential decay applies during inactivity
- Final values are clipped between 0 and 1 for stability

This produces a dynamic, interpretable mastery score that evolves realistically over time.

Conceptually:

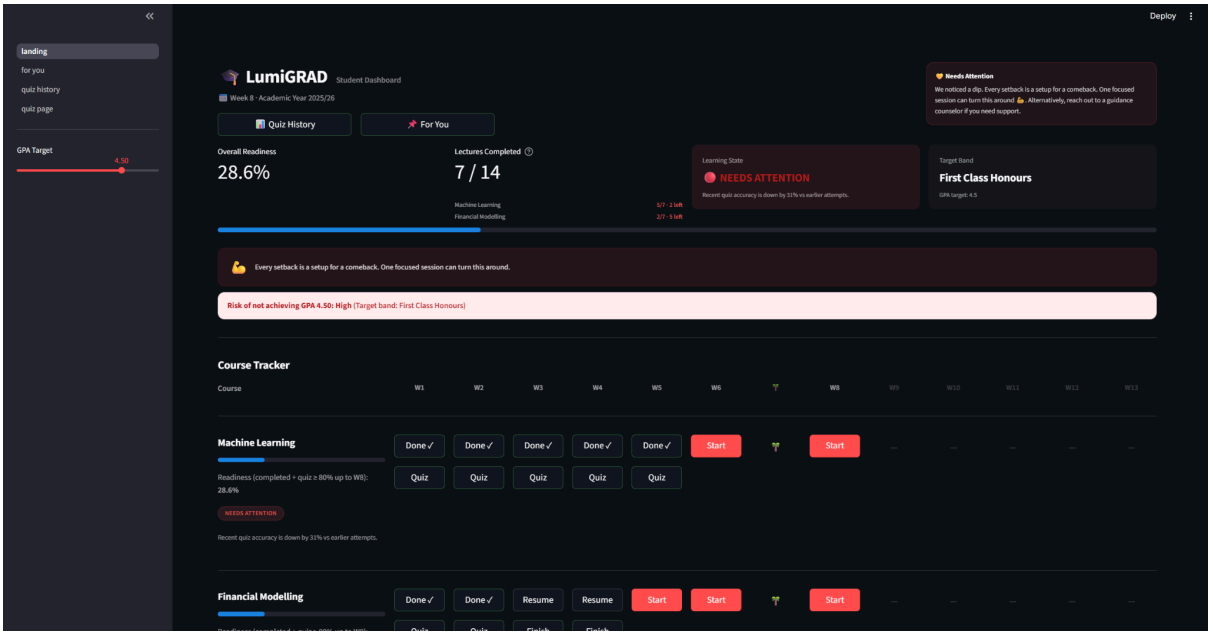
$$\text{Mastery} = w_1(\text{Accuracy}) + w_2(\text{Attempt Confidence}) + w_3(\text{Recency Adjustment})$$

Accuracy reflects correctness, attempt normalisation prevents overconfidence from small samples, and recency weighting ensures that unused knowledge gradually decays. This produces a bounded mastery score between 0 and 1.

Why Rule-Based?
We intentionally chose a transparent, rule-based model instead of black-box predictive machine learning. This ensures interpretability, allows students to understand why their mastery changes, and avoids opaque or potentially unfair academic labeling.

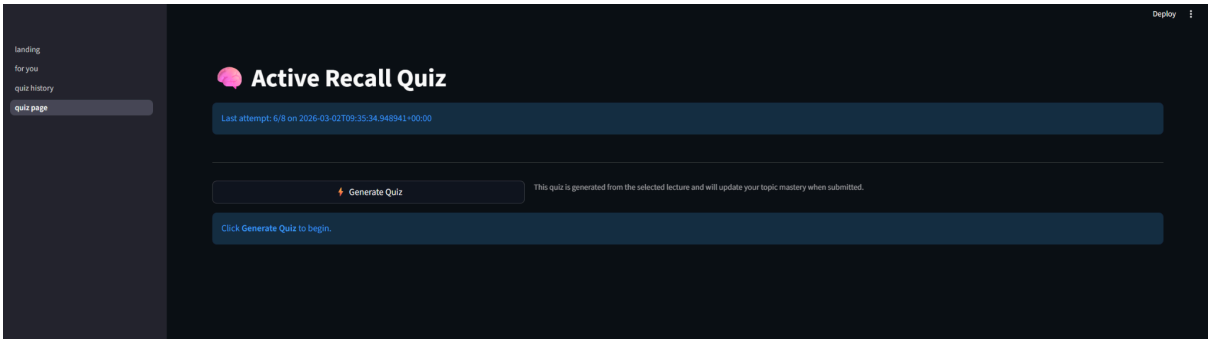
How it looks!

Home Page

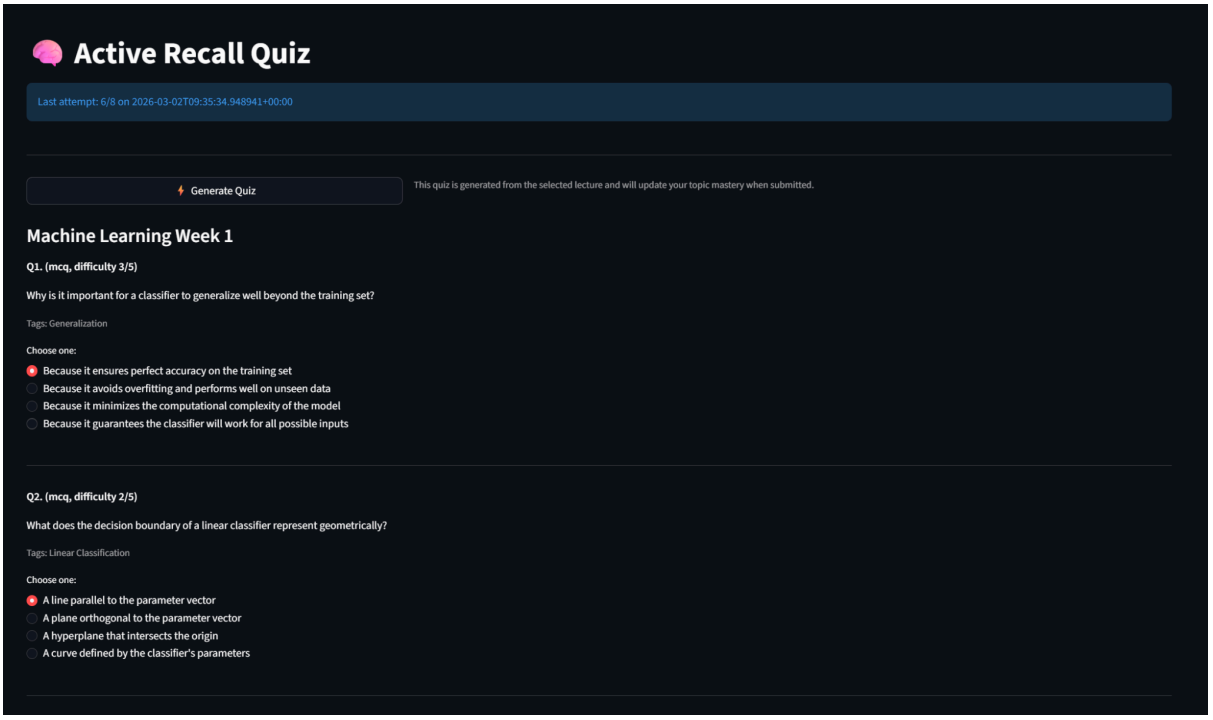


This landing dashboard gives a real-time snapshot of the student's academic trajectory by combining goal-setting, mastery modeling, engagement tracking, and risk estimation into one view. The GPA target defines the benchmark, overall readiness reflects mastery and consistency across modules, lecture completion shows structural progress, and the learning state ("Needs Attention") is derived from recent accuracy trends and engagement patterns. The system highlights the risk of missing the target not to penalise, but to trigger intervention through the planner and reinforcement recommendations, creating a closed feedback loop between assessment, diagnosis, and action.

Quiz Page

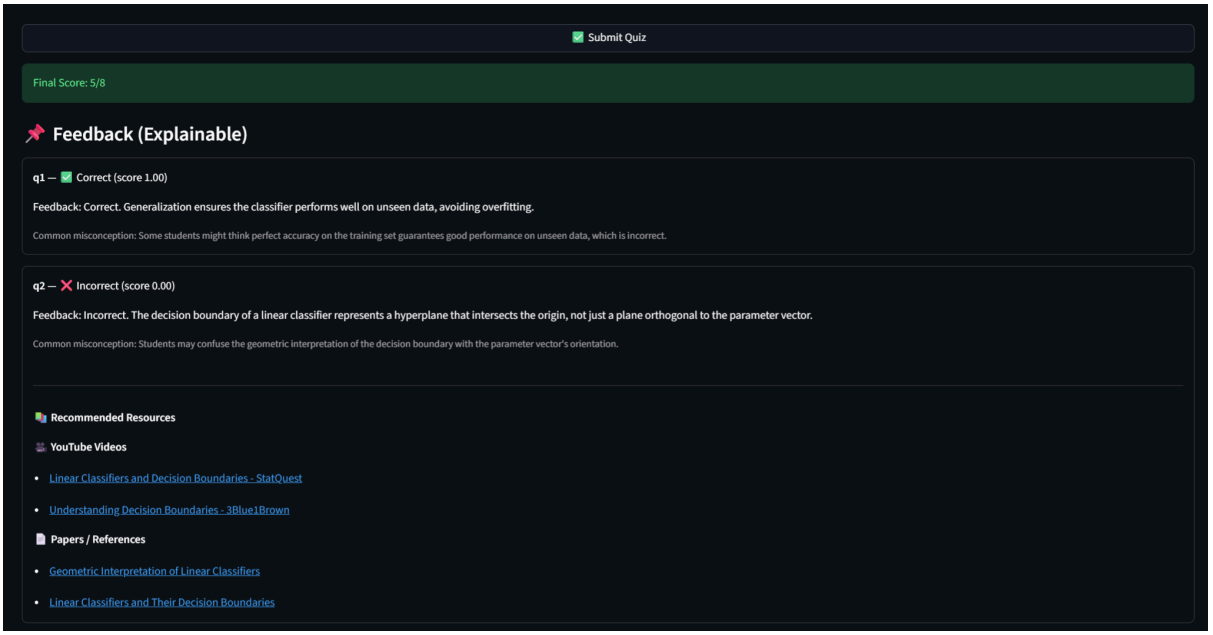


Users are able to see the last attempt of the quiz, if any.
Now we can generate the quiz for a specific lecture on this page. Each question has its own topic tag, which later on can be used to identify topics a student is good and needs more practice in.

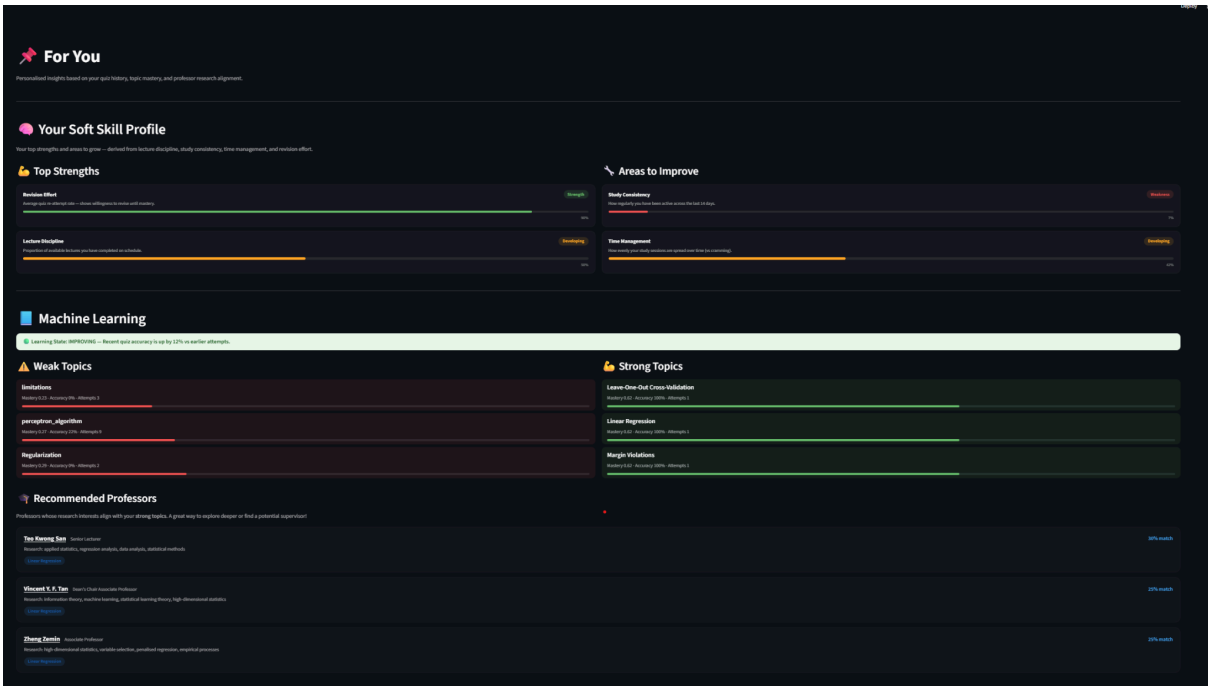


After a quiz is submitted, the GraderAgent evaluates each response and provides structured feedback. For correct answers, it reinforces understanding and highlights common misconceptions to deepen conceptual clarity. For incorrect answers, the ResourceAgent automatically retrieves targeted academic materials related to the specific weak topic, giving

the student an immediate, structured starting point for improvement. This ensures that every assessment directly leads to guided reinforcement rather than ending at a score.



For You Page



Deploy

My Week — Combined Study Schedule

One actionable calendar across **all your modules**, built from your mastery, quiz scores and learning state. Commit your time below and let LumiGRAD plan your week.

Commit to your week

Be honest — a realistic commitment leads to a plan you can actually follow blindly.

Hours available per study day

2.00

Which days can you study this week?

Monday × Tuesday × Wednesday × Thursday × Friday ×

You're committing **10.0 hrs** this week across **5 days**.

Generate My Week

MON
Focus on weakest Machine Learning topic and reinforce Financial Modelling basics.
⏱ 118 min total

MACHINE LEARNING
Review limitations topic and practice related problems
⏱ 60 min
Mastery and accuracy are very low; prioritising weakest topic.
🎯 Improved understanding of limitations and ability to apply concepts.

FINANCIAL MODELLING
Review core principles and work on practice exercises
⏱ 58 min
Maintain stable performance and reinforce understanding of key concepts.
🎯 Enhanced retention of core principles and improved problem-solving skills.

TUE
Strengthen Machine Learning algorithm understanding and deepen Financial Modelling applications.
⏱ 120 min total

MACHINE LEARNING
Study perceptron algorithm and complete related exercises
⏱ 60 min
Accuracy is low; requires focused practice to improve mastery.
🎯 Better grasp of perceptron algorithm and improved accuracy in applications.

FINANCIAL MODELLING
Solve advanced case studies to apply financial concepts
⏱ 60 min
Challenge stable performance with higher-level application tasks.
🎯 Improved ability to apply financial modelling concepts in real-world scenarios.

WED
Address weak Machine Learning topic and refine Financial Modelling forecasting skills.
⏱ 120 min total

MACHINE LEARNING
Work on Regularisation topic and review lecture notes
⏱ 60 min
Mastery and accuracy are very low; needs targeted improvement.
🎯 Enhanced understanding of Regularisation and ability to apply it effectively.

FINANCIAL MODELLING
Practice financial forecasting techniques
⏱ 60 min
Maintain stable performance while exploring forecasting methods.
🎯 Improved forecasting skills and confidence in handling financial data.

THU
Reinforce Machine Learning strengths and explore Financial Modelling risk analysis.
⏱ 118 min total

MACHINE LEARNING
Review strong topics and complete practice quizzes
⏱ 60 min
Reinforce strong topics to ensure retention and boost confidence.
🎯 Solidified understanding of strong topics and improved quiz performance.

FINANCIAL MODELLING
Review risk analysis techniques and apply to case studies
⏱ 58 min
Expand understanding of risk analysis to strengthen overall performance.
🎯 Improved ability to assess risks in financial models.

FRI
Integrate Machine Learning weak topics and assess Financial Modelling proficiency.
⏱ 120 min total

MACHINE LEARNING
Combine weak topics into a comprehensive review session
⏱ 60 min
Consolidate learning of weak topics for better overall mastery.
🎯 Improved integration of weak topics into broader Machine Learning knowledge.

SAT
Rest day

SUN
Rest day

Opportunities For You

Top 2 personalised listings per platform — jobs, events & hackathons — matched to your strongest topics.

Leave-One-Out Cross-Validation · Linear Regression · Margin Violations · Model Selection · Offset Parameter

Generate Opportunities

Jobs

Jobs
LinkedIn

Data Scientist - Predictive Analytics
Grab
📍 Singapore (Hybrid)
This role involves building predictive models using techniques like Linear Regression and Leave-One-Out Cross-Validation.

Machine Learning Engineer
Shopee
📍 Singapore (Hybrid)
The position requires expertise in model selection and tuning offset parameters for scalable machine learning solutions.

See more on Linkedin →

Jobs
Jobstreet

Data Analyst
DBS Bank
📍 Singapore (Onsite)
The role focuses on applying Linear Regression and Margin Violations analysis to financial datasets.

AI Researcher
NTU - Artificial Intelligence Research Centre
📍 Singapore (Onsite)
This research role requires expertise in Leave-One-Out Cross-Validation for evaluating AI models.

See more on Jobstreet →

Jobs
Indeed

Junior Data Scientist
Microsoft
📍 Singapore (Remote)
This role involves implementing model selection strategies for machine learning applications.

ML Engineer - Predictive Models
Google
📍 Singapore (Hybrid)
This position demands skills in offset parameter optimization and cross-validation techniques.

See more on Indeed →

Events

Events
Eventbrite

Events
LinkedIn